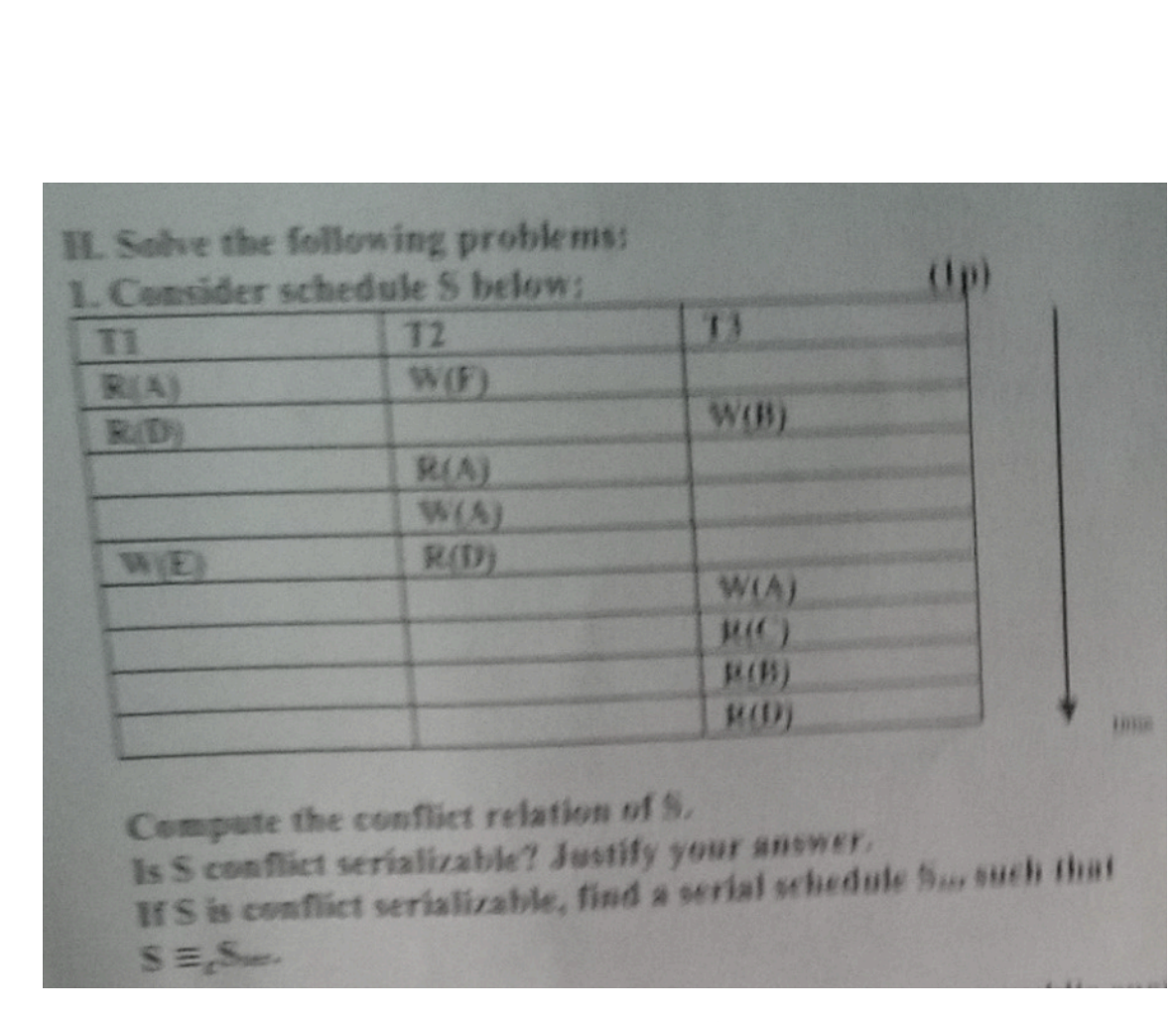
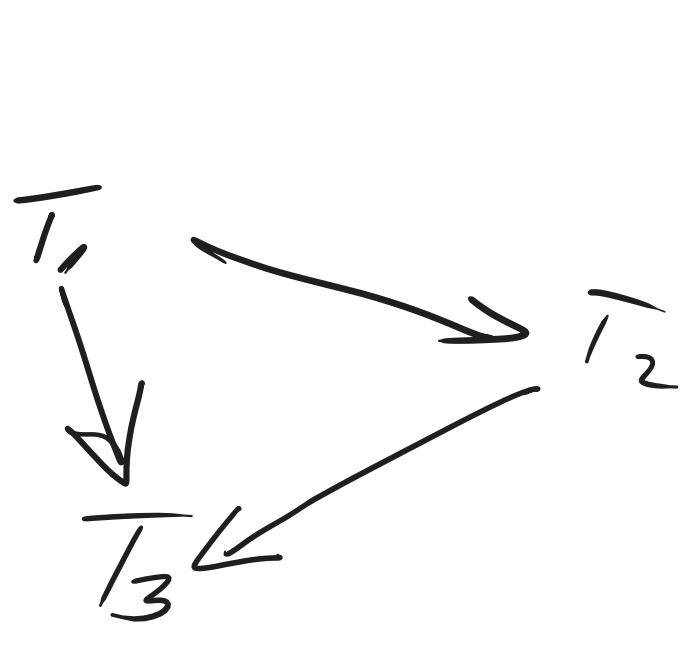


II. Solve the following problems:

1. Consider schedule S below: (1p)

T1	T2	T3
R(A)	W(F)	
R(D)		W(B)
	R(A)	
	W(A)	
W(E)	R(D)	
		W(A)
		R(C)
		R(B)
		R(D)

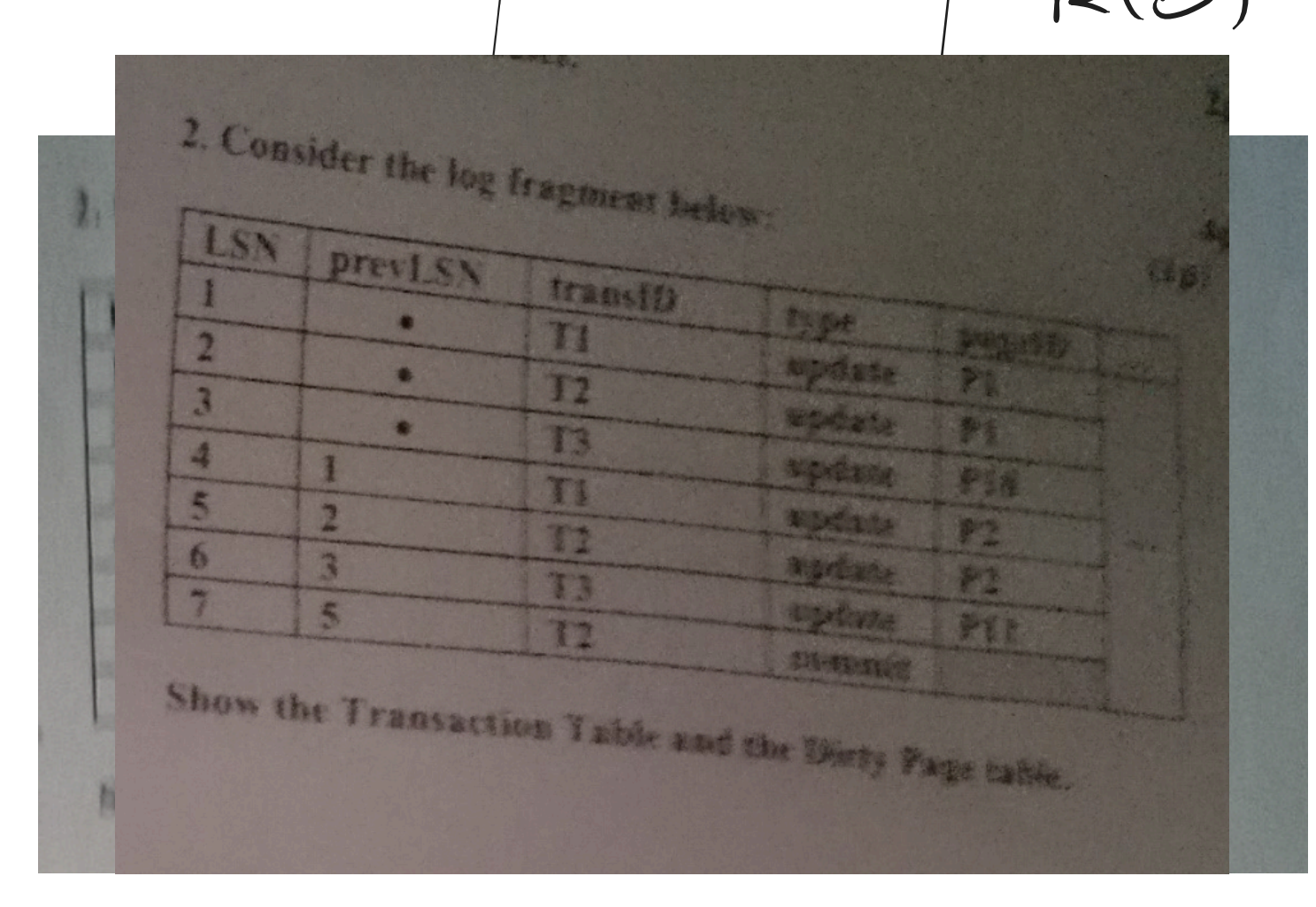
Compute the conflict relation of S.
Is S conflict serializable? Justify your answer.
If S is conflict serializable, find a serial schedule S_{ser} such that $S \equiv_c S_{ser}$.



is CONFLICT SERIALIZABLE

conflict(S) = { (R(T1,A), W(T2,A)), (R(T1,A), W(T3,A)),
(R(T2,A), W(T3,A)), (W(T2,A), W(T3,A)) }

T1	T2	T3
R(A) R(D) W(E)		
	W(F) R(A) W(A) R(D)	
		W(B) W(A) R(C) R(B) R(D)



2. Consider the log fragment below: (1p)

LSN	prevLSN	transID	type	pageID	...
1	•	T1	update	P1	
2	•	T2	update	P1	
3	•	T3	update	P10	...
4	1	T1	update	P2	
5	2	T2	update	P2	
6	3	T3	update	P11	
7	5	T2	commit		

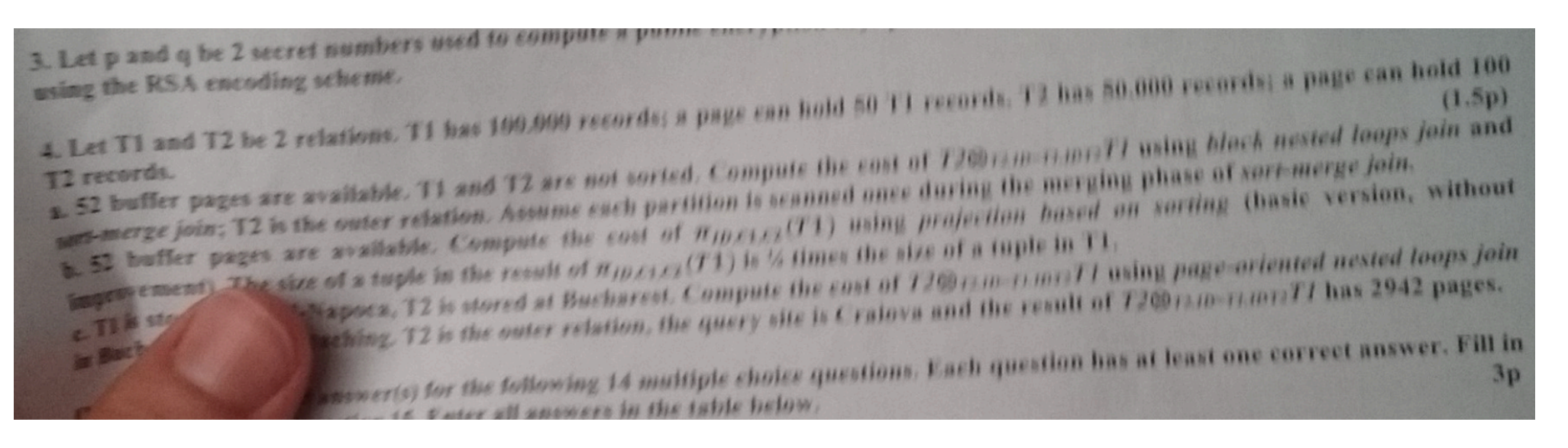
Show the Transaction Table and the Dirty Page table.

Tran table

transID	Status	lastLSN
T1	in progress	4
T2	committed	7
T3	in progress	6

Dirty Page table

page ID	recLSN
P1	1
P10	3
P2	4
P11	6



3. Let p and q be 2 secret numbers used to compute a public
using the RSA encoding scheme.

4. Let T1 and T2 be 2 relations. T1 has 100.000 records; a page can hold 50 T1 records. T2 has 50.000 records; a page can hold 100 T2 records. (1.5p)

a. 52 buffer pages are available. T1 and T2 are not sorted. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *block nested loops join* and *sort-merge join*; T2 is the outer relation. Assume each partition is scanned once during the merging phase of *sort-merge join*.

b. 52 buffer pages are available. Compute the cost of $\pi_{ID,X1,X2}(T1)$ using *projection based on sorting* (basic version, without improvement). The size of a tuple in the result of $\pi_{ID,X1,X2}(T1)$ is ¼ times the size of a tuple in T1.

c. T1 is st Napoca, T2 is stored at Bucharest. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *page-oriented nested loops join* in Buc ching. T2 is the outer relation, the query site is Craiova and the result of $T2 \bowtie_{T2.ID=T1.ID} T1$ has 2942 pages.

answer(s) for the following 14 multiple choice questions. Each question has at least one correct answer. Fill in 3p
Enter all answers in the table below.

T1 : 100 000 r, 50 pe, 2000 pages
T2 : 50 000 r, 100 pe, 500 pgs

a) B = 52

no of blocks = $\left\lceil \frac{2000}{50} \right\rceil = 40$

cost = 500 + 40 * 2000 -> block nested loops join

II sorting

1) compute sorting for both T

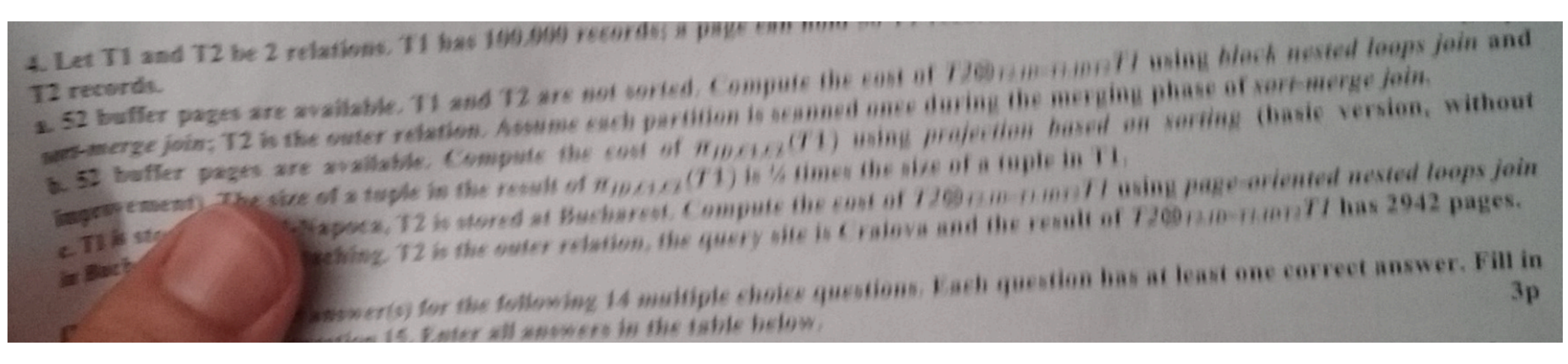
$T_1 : 2 * N * \left(\left\lceil \log_{51} \left\lceil \frac{2000}{52} \right\rceil \right\rceil + 1 \right)$ i/os

$= 4000 \; (3) = 12000$ i/os

$T_2 : 2 * 500 \left(\left\lceil \log_{51} \underbrace{\left\lceil \frac{500}{52} \right\rceil}_{10} \right\rceil + 1 \right)$ i/os

$= 2000$ i/o

cost = 2000 + 500 + 14000 = 16500



4. Let T1 and T2 be 2 relations. T1 has 100.000 records; a page can hold
T2 records.
a. 52 buffer pages are available. T1 and T2 are not sorted. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *block nested loops join* and sort-merge join; T2 is the outer relation. Assume each partition is scanned once during the merging phase of *sort-merge join*.
b. 52 buffer pages are available. Compute the cost of $\pi_{ID,X1,X2}(T1)$ using *projection based on sorting* (basic version, without improvement). The size of a tuple in the result of $\pi_{ID,X1,X2}(T1)$ is ¼ times the size of a tuple in T1.
c. T1 is st Napoca, T2 is stored at Bucharest. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *page-oriented nested loops join* in Buc ching. T2 is the outer relation, the query site is Craiova and the result of $T2 \bowtie_{T2.ID=T1.ID} T1$ has 2942 pages.
answer(s) for the following 14 multiple choice questions. Each question has at least one correct answer. Fill in 3p
Enter all answers in the table below.

b) SCAN T1 : 2000 i/os

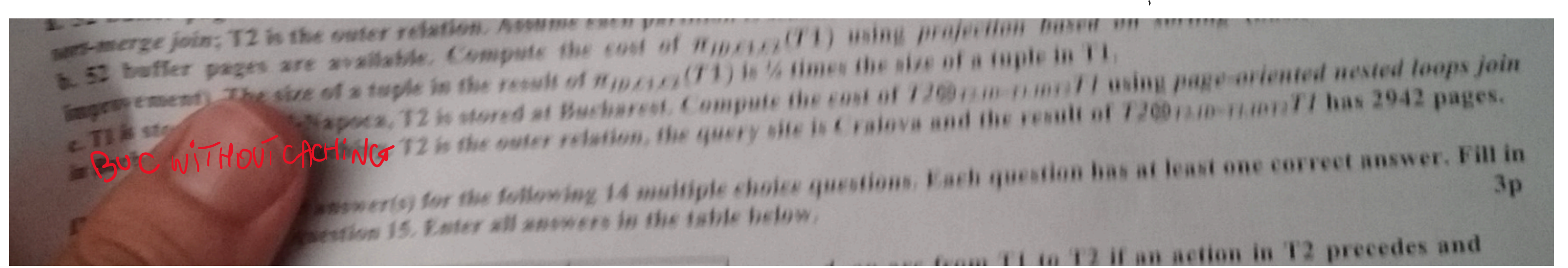
SIZE OF TUPLE T1' : $\frac{2000}{4} = 500$

SORT T1' : $2 * 500 * \left(\left\lceil \log_{51} \left\lceil \frac{500}{52} \right\rceil \right\rceil + 1 \right)$ i/os

$= 2000$

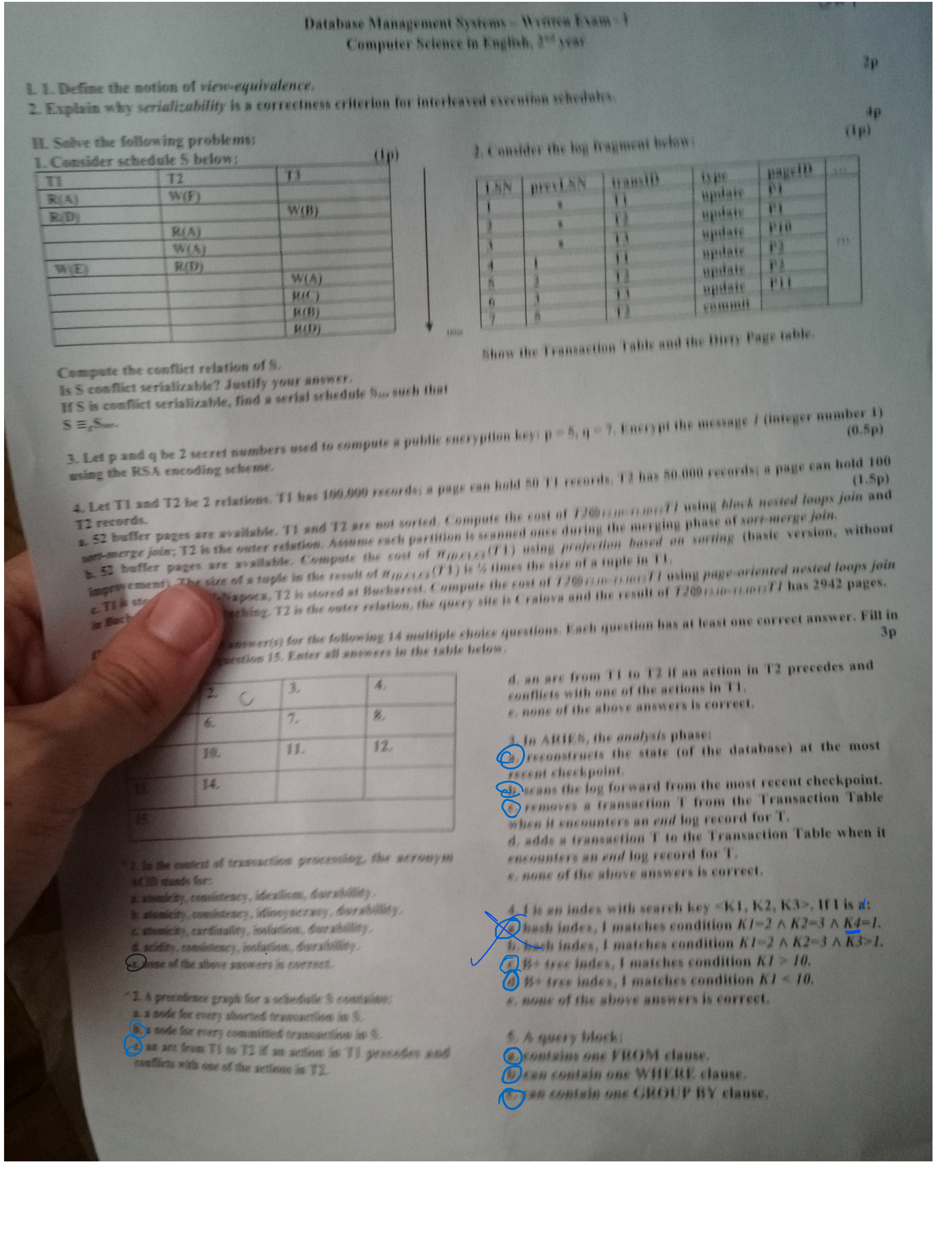
Final SCAN of T1' : 500

cost Total : 5000



sort-merge join; T2 is the outer relation. Assume each partition is scanned once during the merging phase of sort-merge join.
b. 52 buffer pages are available. Compute the cost of $\pi_{ID,X1,X2}(T1)$ using *projection based on sorting* (basic version, without improvement). The size of a tuple in the result of $\pi_{ID,X1,X2}(T1)$ is ¼ times the size of a tuple in T1.
c. T1 is st Napoca, T2 is stored at Bucharest. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *page-oriented nested loops join* in Buc T2 is the outer relation, the query site is Craiova and the result of $T2 \bowtie_{T2.ID=T1.ID} T1$ has 2942 pages.
answer(s) for the following 14 multiple choice questions. Each question has at least one correct answer. Fill in 3p
estion 15. Enter all answers in the table below.

BUG WITHOUT CACHING



Database Management Systems – Written Exam – 1
Computer Science in English, 2nd year

I. 1. Define the notion of *view-equivalence*. 2p
2. Explain why *serializability* is a correctness criterion for interleaved execution schedules. 4p

II. Solve the following problems:
1. Consider schedule S below: (1p)
T1 | T2 | T3
R(A) | W(F) |
R(D) | | W(B)
| R(A) |
| W(A) |
W(E) | R(D) |
| | W(A)
| | R(C)
| | R(B)
| | R(D)
Compute the conflict relation of S.
Is S conflict serializable? Justify your answer.
If S is conflict serializable, find a serial schedule S_{ser} such that $S \equiv_c S_{ser}$.
2. Consider the log fragment below: (1p)
LSN | prevLSN | transID | type | pageID
1 | • | T1 | update | P1
2 | • | T2 | update | P1
3 | • | T3 | update | P10
4 | 1 | T1 | update | P2
5 | 2 | T2 | update | P2
6 | 3 | T3 | update | P11
7 | 5 | T2 | commit |
Show the Transaction Table and the Dirty Page table.

3. Let p and q be 2 secret numbers used to compute a public encryption key: p = 5, q = 7. Encrypt the message *1* (integer number 1) using the RSA encoding scheme. (0.5p)

4. Let T1 and T2 be 2 relations. T1 has 100.000 records; a page can hold 50 T1 records. T2 has 50.000 records; a page can hold 100 T2 records. (1.5p)
a. 52 buffer pages are available. T1 and T2 are not sorted. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *block nested loops join* and *sort-merge join*; T2 is the outer relation. Assume each partition is scanned once during the merging phase of *sort-merge join*.
b. 52 buffer pages are available. Compute the cost of $\pi_{ID,X1,X2}(T1)$ using *projection based on sorting* (basic version, without improvement). The size of a tuple in the result of $\pi_{ID,X1,X2}(T1)$ is ¼ times the size of a tuple in T1.
c. T1 is st Napoca, T2 is stored at Bucharest. Compute the cost of $T2 \bowtie_{T2.ID=T1.ID} T1$ using *page-oriented nested loops join* in Buc ching. T2 is the outer relation, the query site is Craiova and the result of $T2 \bowtie_{T2.ID=T1.ID} T1$ has 2942 pages.
answer(s) for the following 14 multiple choice questions. Each question has at least one correct answer. Fill in 3p
estion 15. Enter all answers in the table below.

	2.	3.	4.
	6.	7.	8.
	10.	11.	12.
13.	14.		
15.			

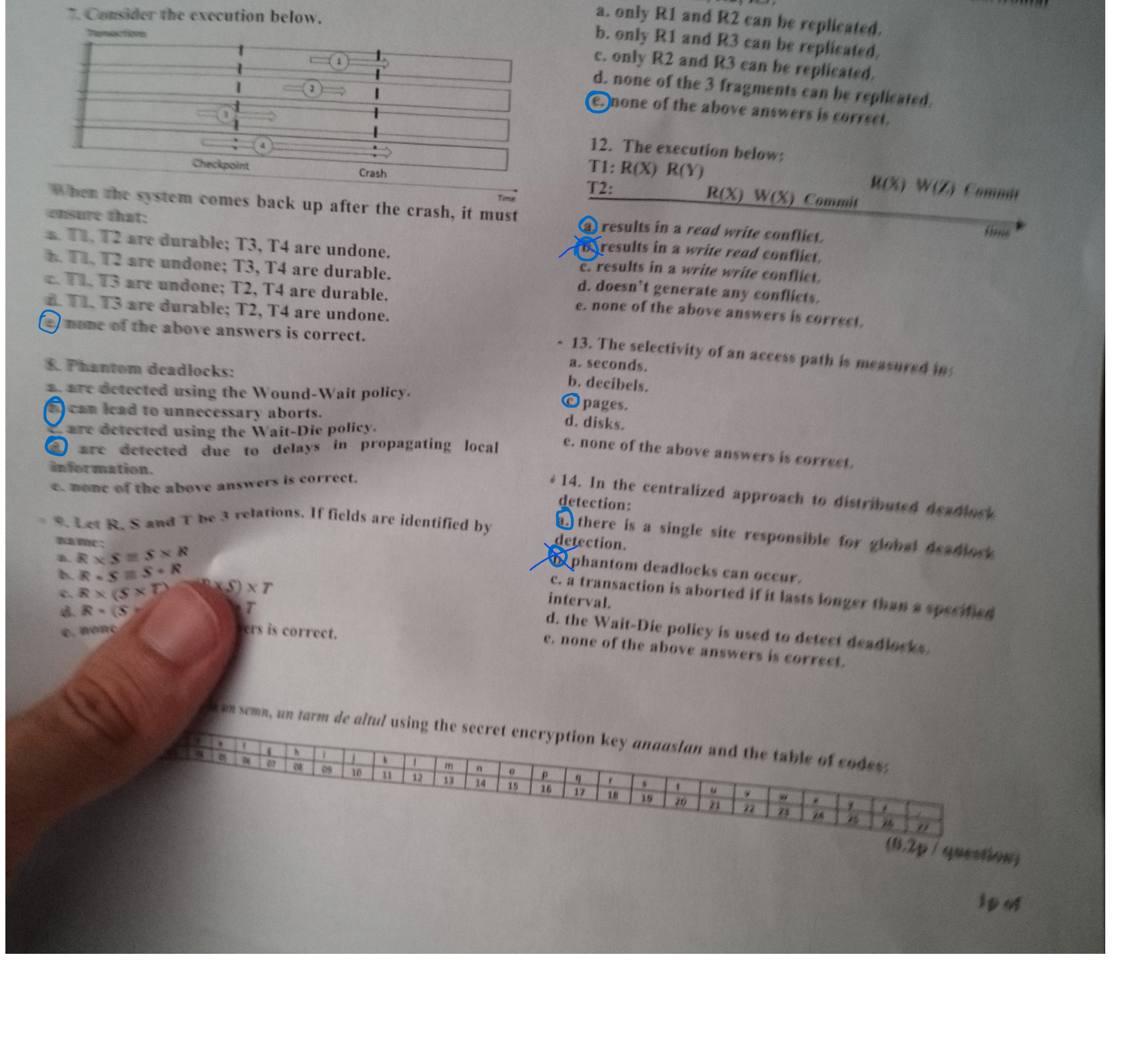
1. In the context of transaction processing, the acronym ACID stands for:
   - a. atomicity, consistency, idealism, durability.
   - b. atomicity, consistency, idiosyncrasy, durability.
   - c. atomicity, cardinality, isolation, durability.
   - d. acidity, consistency, isolation, durability.
   - [✓ e.] none of the above answers is correct.

2. A precedence graph for a schedule S contains:
   - a. a node for every aborted transaction in S.
   - [✓ b.] a node for every committed transaction in S.
   - [✓ c.] an arc from T1 to T2 if an action in T1 precedes and conflicts with one of the actions in T2.
   - d. an arc from T1 to T2 if an action in T2 precedes and conflicts with one of the actions in T1.
   - e. none of the above answers is correct.

3. In ARIES, the *analysis* phase:
   - [✓ a.] reconstructs the state (of the database) at the most recent checkpoint.
   - [✓ b.] scans the log forward from the most recent checkpoint.
   - [✓ c.] removes a transaction T from the Transaction Table when it encounters an *end* log record for T.
   - d. adds a transaction T to the Transaction Table when it encounters an *end* log record for T.
   - e. none of the above answers is correct.

4. I is an index with search key <K1, K2, K3>. If I is a:
   - ☒ a. hash index, I matches condition *K1=2 ∧ K2=3 ∧ K4=1.*
   - b. hash index, I matches condition *K1=2 ∧ K2=3 ∧ K3>1.*
   - [✓ c.] B+ tree index, I matches condition *K1 > 10.*
   - [✓ d.] B+ tree index, I matches condition *K1 < 10.*
   - e. none of the above answers is correct.

5. A query block:
   - [✓ a.] contains one FROM clause.
   - [✓ b.] can contain one WHERE clause.
   - [✓ c.] can contain one GROUP BY clause.



   - [✓ d.] can contain one HAVING clause.
   - e. none of the above answers is correct.

6. The reduction factor for condition *Age = 30*, assuming data is uniformly distributed and there is an index *I* on *Age*, can be estimated by:
   - a. NPages(I)
   - b. 1/NPages(I)
   - [✓ c.] 1/NKeys(I)
   - d. NKeys(I)
   - e. none of the above answers is correct.

7. Consider the execution below.

Transactions
1
2
3
4
Checkpoint
Crash
Time

When the system comes back up after the crash, it must ensure that:
   - a. T1, T2 are durable; T3, T4 are undone.
   - b. T1, T2 are undone; T3, T4 are durable.
   - c. T1, T3 are undone; T2, T4 are durable.
   - d. T1, T3 are durable; T2, T4 are undone.
   - [✓ e.] none of the above answers is correct.

8. Phantom deadlocks:
   - a. are detected using the Wound-Wait policy.
   - [✓ b.] can lead to unnecessary aborts.
   - c. are detected using the Wait-Die policy.
   - [✓ d.] are detected due to delays in propagating local information.
   - e. none of the above answers is correct.

9. Let R, S and T be 3 relations. If fields are identified by name:
   - a. $R \times S \equiv S \times R$
   - b. $R \bowtie S \equiv S \bowtie R$
   - c. $R \times (S \times T)$ … $\times S) \times T$
   - d. $R \bowtie (S$ … T
   - e. none … ers is correct.

10. Regarding concurrency anomalies:
    - a. *dirty reads* can occur under READ COMMITTED.
    - b. *dirty reads* can occur under REPEATABLE READ.
    - [✓ c.] *unrepeatable reads* can occur under READ COMMITTED.
    - d. *unrepeatable reads* can occur under REPEATABLE READ.
    - e. none of the above answers is correct.

11. If a relation R is fragmented into 3 horizontal fragments R1, R2, R3:
    - a. only R1 and R2 can be replicated.
    - b. only R1 and R3 can be replicated.
    - c. only R2 and R3 can be replicated.
    - d. none of the 3 fragments can be replicated.
    - [✓ e.] none of the above answers is correct.

12. The execution below:
    T1: R(X) R(Y) R(X) W(Z) Commit
    T2: R(X) W(X) Commit
    time
    - [✓ a.] results in a *read write* conflict.
    - ☒ b. results in a *write read* conflict.
    - c. results in a *write write* conflict.
    - d. doesn't generate any conflicts.
    - e. none of the above answers is correct.

13. The selectivity of an access path is measured in:
    - a. seconds.
    - b. decibels.
    - [✓ c.] pages.
    - d. disks.
    - e. none of the above answers is correct.

14. In the centralized approach to distributed deadlock detection:
    - [✓ a.] there is a single site responsible for global deadlock detection.
    - ☒ b. phantom deadlocks can occur.
    - c. a transaction is aborted if it lasts longer than a specified interval.
    - d. the Wait-Die policy is used to detect deadlocks.
    - e. none of the above answers is correct.

un semn, un tarm de altul using the secret encryption key *anaaslan* and the table of codes:

(0.2p / question)

1p of